

OS

(Jai Mata di!/ Jai Shivji!)

Module 1 – OS & Process

1. CPU Scheduling
2. CPU Scheduling Algorithms (Advantages & Disadvantages)
3. Multiprocessor Scheduling
4. Real-Time Scheduling
5. Dispatcher
6. Process Control Block (PCB)
7. Threading & Multithreading
8. Inter-Process Communication (IPC)

Module 2 – Synchronization & Deadlock

1. Peterson's Solution
2. Banker's Algorithm
3. Deadlock
4. Deadlock Prevention & Avoidance
5. Producer–Consumer Problem
6. Reader–Writer Problem
7. Dining Philosophers Problem
8. Semaphore
9. Resource Allocation Graph Algorithm

Module 3 – Memory Management

1. Thrashing
2. Paging (and types)
3. Multiprogramming (Fixed & Variable Partitioning)

Module 4 – I/O & File System

1. API (System Calls Interface)

ADDITIONAL IMPORTANT TOPICS

Module 1 –

- Operating System functions & types
- System Calls & System Programs
- Process States & State Diagram
- Context Switching
- Scheduling Criteria (Turnaround, Waiting Time, etc.)
- Types of Threads (User vs Kernel)

Module 2 –

- Critical Section Problem
- Synchronization Hardware (Test-and-Set, Compare-and-Swap)
- Deadlock Characterization (4 Conditions)
- Deadlock Detection & Recovery

Module 3 –

- Segmentation
- Virtual Memory
- Demand Paging
- Page Replacement Algorithms (FIFO, LRU, Optimal)
- Fragmentation (Internal & External)

Module 4 –

- File System Structure
- Directory Structure
- File Allocation Methods
- Free Space Management
- Disk Scheduling Algorithms
- Disk Management
- Kernel I/O Subsystem



*These notes are written by
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